Multimodal Mental Health Emotion Monitoring AI: Master Literature Review & System Blueprint

Curated synthesis of 8 AI-agent literature reviews (ChatGPT, Claude, DeepSeek, Gemini, GLM, Kimi, MiniMax, Qwen) mapped to the project specification

Document: Master Compendium (All-in-One) | **Version:** Release 1.0 (Production-Grade) | **Date:** September 17, 2026 | **Synthesized Sources:** 8 AI Reviews (ChatGPT, Claude, DeepSeek, Gemini, GLM, Kimi, MiniMax, Qwen) + idea.docx

Executive Synthesis & Project Alignment

Project Specification Mapping: AI-Based Mental Health Emotion Monitoring from Facial and Voice Features Using Multimodal AI: face+voice emotion monitoring; CNN/ViT + LSTM/Transformer; MFCC/pitch/spectral features; multimodal fusion; confidence/trend visualization; FER2013, AffectNet, RAVDESS, CREMA-D; real-world robustness; privacy-aware prototype.

This master document represents the complete, exhaustive curation and synthesis of eight independent AI literature reviews (ChatGPT, Claude, DeepSeek, Gemini, GLM, Kimi, MiniMax, Qwen) cross-referenced against the system specification in idea.docx. It provides a publication-grade foundation covering 59 core papers, 15 multimodal datasets, 13 open-source toolkits, 8 critical research gaps, and a fully specified dual-head system architecture blueprint with streaming latency budgets and edge privacy guarantees.

Agent Consensus & Cross-Review Agreement Matrix

To guarantee zero omissions and transparent provenance, every resource in this compendium tracks which of the 8 AI agents cited it. Below is the cross-review consensus distribution across core methodological categories:

Research Category	Key Papers / Models	Consensus Agents	Consensus Key Takeaway
Core Cross-Modal Attention	MulT, MISA, Self-MM	ChatGPT, Claude, Gemini, GLM, Kimi, MiniMax, Qwen	Cross-attention dominates naive concatenation; subspace disentanglement essential for speaker invariance.
Acoustic Foundation Models	Wav2Vec2, HuBERT, WavLM	Claude, DeepSeek, Gemini, GLM, Kimi, MiniMax	Supervised MFCC+LSTM is obsolete; self-supervised speech encoders provide +14% generalization gains.
Facial Vision Transformers	POSTER++, ViT, DAN, MediaPipe	ChatGPT, Claude, DeepSeek, Kimi, MiniMax	Landmark-guided pyramid cross-attention (POSTER++) outperforms frame-level CNNs in-the-wild.
Explainable MLLMs	Emotion-LLaMA, AffectGPT	ChatGPT, Claude, Gemini, GLM, Kimi, MiniMax	Dual-head architecture decouples real-time classification from explainable natural-language reasoning.
Missing Modalities	MMIN, M3S, VAE-Diffusion	ChatGPT, DeepSeek, Kimi, MiniMax, Qwen	Dynamic imputation and latent diffusion prevent catastrophic failure when camera or mic is disabled.
Confidence Calibration	MER-Calibration (Zhang 2024)	ChatGPT, Claude, DeepSeek, Gemini	Post-hoc temperature scaling required to avoid dangerously overconfident clinical predictions.
Clinical Corpora & Transfer	DAIC-WOZ / E-DAIC	Claude, Gemini, MiniMax	Direct bridge from basic emotion categories to clinical depression (PHQ-8) severity scores.

Master Phased Reading Guide (59 Papers)

Priority Tier	Target Scope	Paper Count	Core Representative Papers
Priority P0 (Must Read First)	Architecture Decisions, Core Fusion, Speech/Face Backbones & Clinical Surveys	20 Papers	MulT, MISA, Self-MM, UniMSE, Emotion-LLaMA, Wav2Vec2, HuBERT, POSTER++, DAIC-WOZ, MER-Calibration, MMIN
Priority P1 (Important)	Subspace Disentanglement, Low-Rank Edge Fusion, Missing Modalities & Calibration	20 Papers	MMIM, TFN, LMF, MAG-BERT, DialogueGCN, COSMIC, WavLM, openSMILE, CubeMLP, VAE-Diffusion, M3S
Priority P2 (Specialized)	Conversational Graph Models, Micro-Expressions, Pairwise Alignment & Speech Baselines	14 Papers	DialogueCRN, M2FNet, GS-MCC, MicroEmo, Centering-Hotspots, TITAN, DAN, DepAudioNet, Deep-Depression
Priority P3 (Background Canon)	Historical Reviews, Comprehensive Taxonomies & Benchmark Retrospectives	5 Papers	Baltrusaitis Survey, Rouast Review, Gandhi Survey, Schuller Affect Review, ARM

Complete Literature Review Papers Compendium (59 Papers)

The following catalog presents detailed structured breakdowns for all 59 papers. Each entry details problem formulation, input/output tensors, methodology (how & why), mental health applicability, priority, agent citations, and hyperlinks.

[MulT] Multimodal Transformer for Unaligned Multimodal Language Sequences (2019) — *ACL 2019*

Authors: Yao-Hung Hubert Tsai et al. | **Category:** Core Fusion Architecture
Agent Mentions (7/8): [ChatGPT] [Claude] [Gemini] [GLM] [Kimi] [MiniMax] [Qwen]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [Paper] • [GitHub] • [ArXiv Search] • [Scholar Search]

Problem Solved: Fusing language, vision and audio streams that are unaligned and sampled at different rates, without explicit word-level alignment.

Inputs / Outputs: In: unaligned text+audio+visual sequences. Out: sentiment score / emotion class.

Methodology (How & Why): HOW: directional pairwise cross-modal Transformers (each modality attends to every other) followed by self-attention over concatenated cross-modal outputs. WHY: cross-attention lets one modality discover complementary cues in another (e.g. a frown re-weighting an ambiguous utterance) instead of naive concatenation.

Mental Health Relevance: Reference fusion backbone for the face+voice(+transcript) system; the cross-modal attention pattern is the template for the discriminative head of the blueprint.

[M3ER] M3ER: Multiplicative Multimodal Emotion Recognition Using Facial, Textual, and Speech Cues (2020) — *AAAI 2020*

Authors: Trisha Mittal et al. | **Category:** Core Fusion Architecture
Agent Mentions (3/8): [ChatGPT] [GLM] [Qwen]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [Paper] • [ArXiv Search] • [Scholar Search]

Problem Solved: Noisy or corrupt modalities degrade fusion; standard fusion cannot tell which modality to trust per sample.

Inputs / Outputs: In: face, speech, text cues. Out: emotion class + modality-reliability check.

Methodology (How & Why): HOW: modality-check step filters ineffectual modalities, then multiplicative fusion combines the survivors. WHY: multiplication (vs. addition) lets a confident modality dominate while suppressing corrupted ones.

Mental Health Relevance: Direct template for per-utterance reliability gating in the proposed system (e.g. trust face when audio is noisy).

[MISA] MISA: Modality-Invariant and -Specific Representations for Multimodal Sentiment Analysis (2020) — *ACM MM 2020*

Authors: Devamanyu Hazarika et al. | **Category:** Subspace Disentanglement Fusion
Agent Mentions (6/8): [ChatGPT] [Claude] [Gemini] [GLM] [Kimi] [MiniMax]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [Paper] • [GitHub] • [ArXiv Search] • [Scholar Search]

Problem Solved: Modalities share some affective content but each also carries private noise (identity, articulation, acoustics); entangling them hurts generalization.

Inputs / Outputs: In: text/audio/visual features. Out: sentiment/emotion prediction from fused invariant+specific subspaces.

Methodology (How & Why): HOW: projects each modality into a shared invariant subspace (with similarity loss) plus a private specific subspace (with orthogonality/difference loss), then fuses both. WHY: the invariant space captures consensus affect while specific spaces preserve complementary cues.

Mental Health Relevance: Disentanglement separates speaker identity/channel noise from affect — key for speaker-independent mental-health monitoring.

[MMIM] Improving Multimodal Fusion with Hierarchical Mutual Information Maximization (2021) — *EMNLP 2021*

Authors: Wei Han et al. | **Category:** Information-Theoretic Fusion
Agent Mentions (3/8): [ChatGPT] [GLM] [Kimi]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [Paper] • [GitHub] • [ArXiv Search] • [Scholar Search]

Problem Solved: Fusion layers can silently drop predictive information from weaker modalities during training.

Inputs / Outputs: In: text/audio/visual streams. Out: sentiment prediction with MI-regularized representations.

Methodology (How & Why): HOW: hierarchical mutual-information maximization between input modalities and the fused representation at multiple levels. WHY: explicitly preserving input information prevents the dominant (text) modality from starving audio/visual gradients.

Mental Health Relevance: Useful regularizer when the transcript dominates; keeps facial/vocal signals alive in the fused state used for distress cues.

[Self-MM] Learning Modality-Specific Representations with Self-Supervised Multi-Task Learning (2021) — *AAAI 2021*

Authors: Wei Han et al. | **Category:** Self-Supervised Multi-Task Fusion
Agent Mentions (3/8): [ChatGPT] [GLM] [MiniMax]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [GitHub] • [ArXiv Search] • [Scholar Search]

Problem Solved: Joint multimodal labels are scarce; unimodal annotations are even scarcer, yet each modality needs its own supervision signal.

Inputs / Outputs: In: multimodal sequences. Out: multimodal + auto-generated unimodal predictions.

Methodology (How & Why): HOW: generates pseudo unimodal labels from the multimodal label and trains auxiliary unimodal tasks jointly. WHY: auxiliary tasks force each encoder to be independently competent, improving robustness when a modality is missing at test time.

Mental Health Relevance: Matches the custom-dataset reality: few clinical labels, so self-supervised unimodal objectives squeeze more from every recorded session.

[TFN] Tensor Fusion Network for Multimodal Sentiment Analysis (2017) — *EMNLP 2017*

Authors: Amir Zadeh et al. | **Category:** Tensor Fusion Baseline
Agent Mentions (2/8): [Kimi] [MiniMax]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [GitHub] • [ArXiv Search] • [Scholar Search]

Problem Solved: Early baselines ignored higher-order (tri-modal) interactions between modalities.

Inputs / Outputs: In: language, visual, acoustic embeddings. Out: sentiment score.

Methodology (How & Why): HOW: outer product of modality embeddings (with bias) to model unimodal, bimodal and trimodal interactions explicitly. WHY: multiplicative interactions capture joint dynamics a concatenation layer must learn slowly.

Mental Health Relevance: Classic baseline to beat in the thesis comparison table; too heavy/slow for the real-time head but a standard reference point.

[LMF] Efficient Low-rank Multimodal Fusion with Modality-Specific Factors (2018) — *ACL 2018*

Authors: Zhilin Liu et al. | **Category:** Tensor Fusion Baseline
Agent Mentions (2/8): [Kimi] [MiniMax]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [Paper] • [ArXiv Search] • [Scholar Search]

Problem Solved: TFN's full tensor scales exponentially with modalities and hidden size.

Inputs / Outputs: In: modality embeddings. Out: sentiment prediction via low-rank factors.

Methodology (How & Why): HOW: low-rank decomposition of the fusion tensor into modality-specific factors. WHY: same expressive tri-modal interactions at a fraction of the parameters — deployable on edge.

Mental Health Relevance: Candidate lightweight fusion for on-device inference where transformer fusion is too heavy.

[MFN] Memory Fusion Network for Multi-view Sequential Learning (2018) — *AAAI 2018*

Authors: Amir Zadeh et al. | **Category:** Sequential Fusion Baseline
Agent Mentions (3/8): [GLM] [Kimi] [MiniMax]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Emotion unfolds over time and modalities interact across timesteps, not just within one utterance.

Inputs / Outputs: In: multi-view temporal sequences. Out: per-sequence affect prediction.

Methodology (How & Why): HOW: per-modality LSTMs plus a delta-memory attention that tracks cross-view interactions over time. WHY: an explicit memory of how modalities influenced each other historically.

Mental Health Relevance: Historical precedent for the continuous single-user temporal modeling the project needs (speaking/silent state transitions).

[MAG-BERT] Integrating Multimodal Information in Large Pretrained Transformers (MAG-BERT) (2020) — *ACL 2020*

Authors: Wasifur Rahman et al. | **Category:** Transformer Fusion
Agent Mentions (2/8): [Kimi] [MiniMax]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [Paper] • [ArXiv Search] • [Scholar Search]

Problem Solved: Fine-tuning BERT on text alone discards acoustic/visual cues; full multimodal retraining is expensive.

Inputs / Outputs: In: text tokens + nonverbal (audio/visual) features. Out: sentiment/emotion label.

Methodology (How & Why): HOW: Multimodal Adaptation Gate shifts BERT's internal representations conditioned on audio/visual features. WHY: cheaply injects nonverbal context into a frozen-strong language model.

Mental Health Relevance: Pattern for conditioning a strong pretrained backbone (e.g. wav2vec2/BERT) on the weaker face/voice stream without full retraining.

[M2FNet] M2FNet: Multi-modal Fusion Network for Emotion Recognition in Conversation (2022) — *CVPR Workshops 2022*

Authors: Vishal Chudasama et al. | **Category:** Conversational Fusion
Agent Mentions (1/8): [ChatGPT]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Conversational emotion needs both modality fusion and dialogue context in one model.

Inputs / Outputs: In: face, audio, text across dialogue turns. Out: per-utterance emotion.

Methodology (How & Why): HOW: modality-specific encoders plus scene/context modelling fused for conversation-level prediction. WHY: joint modelling avoids cascading errors of separate context/fusion stages.

Mental Health Relevance: Relevant context modeller, though the project is single-user rather than multi-party dialogue.

[Noise-Resistant-MMT] Noise-Resistant Multimodal Transformer for Emotion Recognition (2023) — *IEEE TAFFC 2023*

Authors: H. Li et al. | **Category:** Robust Fusion
Agent Mentions (2/8): [ChatGPT] [MiniMax]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Real-world audio/visual streams are noisy; vanilla transformers propagate that noise into the fused state.

Inputs / Outputs: In: noisy multimodal sequences. Out: noise-robust emotion prediction.

Methodology (How & Why): HOW: noise-aware attention/filtering inside the multimodal transformer. WHY: explicitly down-weights corrupted frames/segments instead of averaging them in.

Mental Health Relevance: Directly supports the 'uncontrolled conditions' evaluation: lighting variation, background noise, sensor artefacts.

[Fusing-Pairwise] Fusing Pairwise Modalities for Emotion Recognition in Conversations (2023) — *INTERSPEECH 2023*

Authors: Guo et al. | **Category:** Conversational Fusion
Agent Mentions (1/8): [ChatGPT]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Trimodal joint fusion can blur which modality pair actually carries the signal.

Inputs / Outputs: In: pairwise modality combinations across turns. Out: conversational emotion labels.

Methodology (How & Why): HOW: fuses modalities pairwise before a final aggregation. WHY: pairwise interactions are easier to learn and diagnose than one-shot trimodal mixing.

Mental Health Relevance: Diagnostic fusion style: per-pair confidences map naturally onto the dashboard's per-modality confidence display.

[MER-Calibration] Multimodal Emotion Recognition Calibration in Conversations (2024) — *EMNLP 2024*

Authors: Zhang et al. | **Category:** Calibration & Reliability
Agent Mentions (4/8): [ChatGPT] [Claude] [DeepSeek] [Gemini]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: MER models are overconfident; reported confidence scores do not match empirical accuracy.

Inputs / Outputs: In: trained MER model + validation data. Out: calibrated confidence estimates.

Methodology (How & Why): HOW: post-hoc/online calibration of multimodal confidence. WHY: trustworthy confidence is a safety requirement, not a luxury, in health-adjacent systems.

Mental Health Relevance: Foundational for the confidence-score display in the app: uncalibrated scores would mislead users about their own state.

[Self-Attention-Incomplete] Self-Attention Fusion for Audiovisual Emotion Recognition with Incomplete Data (2021) — *IEEE TAFFC 2021*

Authors: Radhakrishnan et al. / Praveen et al. | **Category:** Missing Modalities / Robustness
Agent Mentions (1/8): [ChatGPT]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Audiovisual samples with a missing or corrupted stream must still be classified.

Inputs / Outputs: In: incomplete audio-visual data. Out: emotion label robust to missing input.

Methodology (How & Why): HOW: self-attention pooling that dynamically re-weights available frames/modalities. WHY: attention naturally handles variable-length, partially observed inputs.

Mental Health Relevance: Baseline missing-modality strategy for camera-off / mic-muted operation modes.

[MMIN] Missing Modality Imagination Network for Emotion Recognition with Incomplete Audio-Visual Data (2021) — *ACL 2021*

Authors: Jiandian Zhao et al. | **Category:** Missing Modalities / Robustness
Agent Mentions (1/8): [Kimi]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: At test time a whole modality can vanish; the model must imagine it from what remains.

Inputs / Outputs: In: modality-incomplete test samples (full modalities at train). Out: emotion prediction via imagined features.

Methodology (How & Why): HOW: learns to hallucinate missing-modality features with cycle-consistency constraints. WHY: reconstructing features (not raw signals) keeps classification alive when sensors drop.

Mental Health Relevance: Core technique for graceful degradation: muted mic or covered camera still yields a prediction with honest uncertainty.

[IF-MMIN] Information-Theoretic Feature Learning for Missing Modality Imagination (2022) — *arXiv / IEEE Access 2022*

Authors: Zhao / Liu et al. | **Category:** Missing Modalities / Robustness
Agent Mentions (1/8): [Kimi]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: MMIN-style imagination can hallucinate plausible-but-wrong features without information-theoretic grounding.

Inputs / Outputs: In: incomplete multimodal input. Out: imagined features + emotion label.

Methodology (How & Why): HOW: mutual-information objectives constrain the imagination module. WHY: guarantees the hallucinated features preserve label-relevant information.

Mental Health Relevance: Safer imagination: in health monitoring, a wrong-but-confident hallucination is worse than an honest 'modality missing' flag.

[M3S] M3S: Missing Modality meets Meta-Sampling (2022) — *AACL 2022*

Authors: Various | **Category:** Missing Modalities / Robustness
Agent Mentions (1/8): [MiniMax]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Training rarely covers the combinatorial space of missing-modality patterns seen at deployment.

Inputs / Outputs: In: multimodal training data. Out: model robust across sampled missingness patterns.

Methodology (How & Why): HOW: meta-sampling simulates diverse missing-modality configurations during training. WHY: explicit coverage of the missingness distribution instead of hoping dropout generalizes.

Mental Health Relevance: Training recipe for the project's robustness evaluation (camera-off, mic-off, intermittent drops).

[VAE-Diffusion-MER] HVDER: VAE-Integrated Multiscale Generative Diffusion Modeling for Incomplete Multimodal Emotion Recognition (2026) — *IEEE 2026*

Authors: Various | **Category:** Generative Modality Imputation
Agent Mentions (1/8): [DeepSeek]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: High missing rates (up to 70%) destroy discriminative fusion; prior imagination methods collapse under severe missingness.

Inputs / Outputs: In: modality-incomplete sequences. Out: completed latent representations + emotion label.

Methodology (How & Why): HOW: modality-specific VAEs project to a unified latent space, then a two-stage coarse-to-fine conditional diffusion completes missing latents (76.8% acc on CMU-MOSEI). WHY: generative completion in compact latent space is more stable than raw-signal inpainting.

Mental Health Relevance: State-of-the-art missing-modality handler, but needs streaming validation before deployment in the real-time pipeline.

[FDFNet] Feature Disentanglement and Fusion Network for Multimodal Emotion Recognition (2023) — *IEEE / ACM MM 2023*

Authors: Various | **Category:** Subspace Disentanglement
Agent Mentions (1/8): [Gemini]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Entangled shared/private features limit both interpretability and robustness.

Inputs / Outputs: In: multimodal features. Out: disentangled + fused emotion prediction.

Methodology (How & Why): HOW: explicit disentanglement branches with fusion over separated factors. WHY: cleaner factors give both accuracy and per-factor explanations.

Mental Health Relevance: Disentangled factors can be surfaced as explanations ('voice drove this prediction, face was neutral').

[EMER] Ensemble Multimodal Emotion Recognition with Dynamic Modality Weighting (2024) — *IEEE TAFFC 2024*

Authors: Various | **Category:** Adaptive Fusion
Agent Mentions (2/8): [ChatGPT] [Gemini]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Static fusion weights fail when modality reliability changes utterance to utterance.

Inputs / Outputs: In: multimodal streams. Out: ensemble prediction with dynamic weights.

Methodology (How & Why): HOW: ensemble of modality experts with learned per-sample weighting. WHY: adapts to instantaneous sensor quality.

Mental Health Relevance: Ensemble weights double as the dashboard's per-modality confidence bars.

[UniMSE] UniMSE: Towards Unified Multimodal Sentiment Analysis and Emotion Recognition (2022) — *EMNLP 2022*

Authors: Guimin Hu et al. | **Category:** Unified Affective Framework
Agent Mentions (2/8): [ChatGPT] [Gemini]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [Paper] • [ArXiv Search] • [Scholar Search]

Problem Solved: Sentiment analysis and emotion recognition are trained as separate tasks despite sharing representations.

Inputs / Outputs: In: multimodal utterance. Out: joint sentiment + emotion predictions.

Methodology (How & Why): HOW: unified encoder with task-specific heads and cross-task pretraining (T5-based). WHY: shared affective knowledge transfers between intensity (sentiment) and category (emotion).

Mental Health Relevance: Blueprint for the dual head: one backbone serving both categorical emotions and dimensional intensity for trend tracking.

[CubeMLP] CubeMLP: An MLP-based Model for Multimodal Sentiment Analysis and Depression Detection (2022) — *ACM MM 2022*

Authors: Hao Sun et al. | **Category:** Low-Latency Architecture
Agent Mentions (1/8): [Gemini]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Transformer fusion is too slow/memory-heavy for edge and real-time use, including depression screening.

Inputs / Outputs: In: multimodal features. Out: sentiment / depression prediction.

Methodology (How & Why): HOW: pure-MLP mixing across modality, sequence and channel axes (a 'cube' of mixers). WHY: MLP mixing is linear-time and tiny, yet competitive — ideal latency/accuracy trade-off.

Mental Health Relevance: Candidate on-device fusion for the edge-privacy requirement; also one of few MER-family models evaluated on depression detection.

[DialogueRNN] DialogueRNN: An Attentive RNN for Emotion Detection in Conversation (2019) — *AAAI 2019*

Authors: Navonil Majumder et al. | **Category:** Conversational Context
Agent Mentions (4/8): [ChatGPT] [GLM] [Kimi] [MiniMax]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [GitHub] • [ArXiv Search] • [Scholar Search]

Problem Solved: Utterance emotion depends on speaker state and conversational history, not just the current utterance.

Inputs / Outputs: In: sequence of utterances with speaker IDs. Out: per-utterance emotion.

Methodology (How & Why): HOW: party/state/context GRUs track each speaker plus global context with attention. WHY: separates 'who I am being' (speaker state) from 'what was said' (context).

Mental Health Relevance: Adapt the speaker-state idea to a single user: a persistent user-state GRU across utterances/silences models mood drift.

[DialogueGCN] DialogueGCN: A Graph Convolutional Network for Emotion Recognition in Conversation (2019) — *EMNLP-IJCNLP 2019*

Authors: Deepanway Ghosal et al. | **Category:** Conversational Context
Agent Mentions (3/8): [ChatGPT] [Kimi] [MiniMax]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [Paper] • [ArXiv Search] • [Scholar Search]

Problem Solved: RNNs propagate context only sequentially, missing long-range speaker dependencies.

Inputs / Outputs: In: dialogue utterance graph. Out: per-utterance emotion.

Methodology (How & Why): HOW: builds a speaker-aware graph over utterances and applies graph convolution. WHY: graphs capture long-range, non-sequential influence (an early remark coloring a much later reply).

Mental Health Relevance: Graph-over-session view supports session-level summaries ('this session trended negative after minute 6').

[COSMIC] COSMIC: COmmonSense knowledge for eMotion Identification in Conversations (2020) — *COLING 2020*

Authors: Deepanway Ghosal et al. | **Category:** Conversational Context
Agent Mentions (3/8): [ChatGPT] [Kimi] [MiniMax]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [GitHub] • [ArXiv Search] • [Scholar Search]

Problem Solved: Surface text misses the mental-state layer (intent, reaction, effect) behind emotional expression.

Inputs / Outputs: In: dialogue with commonsense (COMET) features. Out: per-utterance emotion.

Methodology (How & Why): HOW: injects COMET commonsense inferences (speaker intent, listener effect) into the context model. WHY: emotions follow from appraised events, not just words.

Mental Health Relevance: Commonsense layer is the seed of explainability: 'flat prosody + withdrawn wording suggests low-arousal negative state'.

[DialogueCRN] DialogueCRN: Contextual Reasoning Networks for Emotion Recognition in Conversation (2021) — *ACL 2021*

Authors: Dou Hu et al. | **Category:** Conversational Context
Agent Mentions (2/8): [ChatGPT] [Gemini]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Shallow context aggregation cannot perform multi-hop reasoning over dialogue situations.

Inputs / Outputs: In: conversational utterances. Out: emotion via retrieved situations + reasoning.

Methodology (How & Why): HOW: cognitive situation retrieval plus multi-turn reasoning over retrieved contexts. WHY: explicit reasoning steps are auditable, unlike black-box recurrence.

Mental Health Relevance: Auditable reasoning steps suit the explainable-MLLM head: each claim in an explanation can cite its evidence turn.

[GS-MCC] GS-MCC: Graph Structure Learning with Multi-Context Constraints for ERC (2025) — *AAAI 2025*

Authors: Ai et al. | **Category:** Conversational Context
Agent Mentions (1/8): [Kimi]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Hand-designed dialogue graphs may encode the wrong structure for a given conversation.

Inputs / Outputs: In: utterances with multi-context features. Out: emotion with a learned graph.

Methodology (How & Why): HOW: learns the graph structure itself under multi-context constraints. WHY: data-driven edges adapt to single-user monologue as well as multi-party dialogue.

Mental Health Relevance: Structure learning transfers to the single-user setting where no speaker graph exists a priori.

[Emotion-LLaMA] Emotion-LLaMA: Multimodal Emotion Recognition and Reasoning with Instruction Tuning (2024) — *NeurIPS 2024 / arXiv 2024*

Authors: Zebang Cheng et al. | **Category:** Multimodal LLM / Explainability
Agent Mentions (5/8): [ChatGPT] [Claude] [Gemini] [Kimi] [MiniMax]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [Paper] • [GitHub] • [ArXiv Search] • [Scholar Search]

Problem Solved: Classifiers output labels without reasons; clinicians and users need grounded explanations.

Inputs / Outputs: In: audio+visual (+text) clips. Out: emotion label + natural-language rationale.

Methodology (How & Why): HOW: instruction-tunes an MLLM on emotion-recognition-and-reasoning data. WHY: instruction tuning aligns generation with affective evidence instead of free hallucination.

Mental Health Relevance: Direct template for the explainable head of the blueprint: labels from the discriminative head, rationales from an instruction-tuned MLLM.

[MERBench] MERBench: A Unified Evaluation Benchmark for Multimodal Emotion Recognition in the Wild (2024) — *ACM MM 2024 / IEEE TAFFC 2024*

Authors: Zheng Lian et al. | **Category:** Evaluation Benchmark & Toolkit
Agent Mentions (2/8): [ChatGPT] [MiniMax]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [GitHub] • [ArXiv Search] • [Scholar Search]

Problem Solved: Inconsistent splits, metrics and preprocessing make MER results incomparable across papers.

Inputs / Outputs: In: any MER model. Out: standardized benchmark scores.

Methodology (How & Why): HOW: unified data pipeline, fixed splits, and standard metrics across corpora. WHY: comparability is a precondition for claiming progress.

Mental Health Relevance: Adopt MERBench protocol for the thesis ablations so single-modal vs multimodal comparisons are credible.

[AffectGPT] AffectGPT: A New Dataset, Model, and Benchmark for Multimodal Emotion Recognition and Reasoning (2024/2025) — *ICML 2025 / arXiv 2024*

Authors: Various | **Category:** Multimodal LLM / Affective Reasoning
Agent Mentions (4/8): [Claude] [Gemini] [GLM] [MiniMax]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [GitHub] • [Toolkit] • [ArXiv Search] • [Scholar Search]

Problem Solved: No dataset jointly supports emotion recognition AND reasoning evaluation for MLLMs.

Inputs / Outputs: In: multimodal clips. Out: labels + reasoning traces scored on a dedicated benchmark.

Methodology (How & Why): HOW: new dataset with reasoning annotations plus an MLLM baseline and benchmark suite. WHY: reasoning quality must be measured, not demoed.

Mental Health Relevance: Reasoning benchmark informs how to evaluate the explainable head beyond label accuracy (faithfulness, grounding).

[EMOE] EMOE: Modality-Specific Enhanced Dynamic Emotion Experts (2024) — *ACM MM 2024*

Authors: Various | **Category:** Mixture of Experts MER
Agent Mentions (1/8): [ChatGPT]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: One shared fusion network cannot specialize in the distinct statistics of each modality.

Inputs / Outputs: In: multimodal streams. Out: mixture-of-experts emotion prediction.

Methodology (How & Why): HOW: modality-specific expert networks with a dynamic router. WHY: specialization plus conditional compute (only relevant experts fire).

Mental Health Relevance: Router activations explain 'which signal mattered'; conditional compute helps real-time budgets.

[M-SENA] M-SENA: An Integrated Platform for Multimodal Sentiment Analysis and Emotion Recognition (2023) — *ACL 2023 System Demo*

Authors: Various | **Category:** System Platform & Toolkit
Agent Mentions (1/8): [GLM]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Researchers reimplement feature extraction, training and evaluation glue for every experiment.

Inputs / Outputs: In: raw multimodal corpora. Out: reproducible baselines and analyses.

Methodology (How & Why): HOW: integrated platform covering data, features, models and evaluation. WHY: standardization accelerates honest ablations.

Mental Health Relevance: Use as the experiment harness for the thesis: one platform for face-only, voice-only and fused runs.

[Centering-Emotion-Hotspots] Centering Emotion Hotspots: Multimodal Local-Global Fusion for Conversation-Level Emotion Recognition (2024) — *ACM MM 2024*

Authors: Various | **Category:** Local-Global Attention
Agent Mentions (1/8): [Qwen]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Emotion concentrates in short 'hotspot' regions; global pooling dilutes them.

Inputs / Outputs: In: conversation-level multimodal streams. Out: conversation emotion with hotspot localization.

Methodology (How & Why): HOW: local-global fusion that centers on detected emotional hotspots. WHY: preserves brief but decisive cues (a tremor, a micro-expression) inside long sessions.

Mental Health Relevance: Hotspot localization maps to session timelines in the dashboard ('distress hotspot at 04:12').

[MicroEmo] MicroEmo: Time-Sensitive Multimodal Language Model for Micro-Expression Recognition (2024) — *ECCV 2024*

Authors: Various | **Category:** Micro-Expression / Temporal
Agent Mentions (5/8): [Qwen] [Claude] [DeepSeek] [Gemini] [Kimi]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Brief involuntary micro-expressions leak concealed affect but are missed by frame classifiers.

Inputs / Outputs: In: high-frame-rate face video (+audio). Out: micro-expression-aware emotion.

Methodology (How & Why): HOW: time-sensitive MLLM modelling of fleeting facial dynamics. WHY: explicit temporal sensitivity at sub-second scale.

Mental Health Relevance: Long-term extension: concealed-distress detection; note frame-rate and compute costs before promising it.

[EmoFusion] EmoFusion: Anchor-Guided Multimodal Emotion Recognition With Cross-Modal Attention (2023) — *IEEE TAFFC 2023*

Authors: Various | **Category:** Anchor-Guided Attention
Agent Mentions (1/8): [Qwen]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Cross-modal attention without guidance can latch onto spurious correlations.

Inputs / Outputs: In: multimodal utterance. Out: anchored fused prediction.

Methodology (How & Why): HOW: anchor-guided attention constrains cross-modal alignment to reliable reference points. WHY: anchors stabilize training and improve interpretability of attention maps.

Mental Health Relevance: Attention maps become presentable evidence overlays in the monitoring report.

[TITAN] TITAN: Measuring Latent Speech-Text Alignment for Spoken Language Understanding (2023) — *ICASSP 2023*

Authors: Various | **Category:** Acoustic-Textual Alignment
Agent Mentions (1/8): [Qwen]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Speech-text misalignment (ASR error, sarcasm, disfluency) silently corrupts trimodal fusion.

Inputs / Outputs: In: speech + transcript. Out: alignment quality measure + downstream label.

Methodology (How & Why): HOW: quantifies latent alignment between acoustic and textual streams. WHY: a measurable alignment score gates how much the text stream is trusted.

Mental Health Relevance: Sarcasm/neutral-words-with-sad-voice cases: low alignment triggers voice-priority fallback, a clinically important behavior.

[KAMUH] KAMUH: A Real-Time Online Audiovisual Emotion Recognition System (2022) — *ICMI 2022*

Authors: Rasendrasoa et al. | **Category:** Real-Time / Streaming MER
Agent Mentions (1/8): [Kimi]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Benchmark MER assumes pre-segmented utterances; live systems must segment, buffer and predict online.

Inputs / Outputs: In: live audiovisual stream. Out: online emotion predictions.

Methodology (How & Why): HOW: streaming architecture with online segmentation and incremental inference. WHY: proves utterance-level MER can run under latency constraints.

Mental Health Relevance: Closest deployment reference for the app's streaming pipeline: 3-second audio windows + per-frame face classification pattern.

[POSTER++] POSTER++: A Simpler and Stronger Facial Expression Recognition Network (2023) — *ICCV 2023*

Authors: Jiawei Mao et al. | **Category:** Facial Expression Backbone
Agent Mentions (2/8): [Claude] [Kimi]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [GitHub] • [Paper] • [ArXiv Search] • [Scholar Search]

Problem Solved: Two-stream FER (image + landmarks) needs simpler cross-stream interaction without losing accuracy.

Inputs / Outputs: In: face image + landmarks. Out: expression class.

Methodology (How & Why): HOW: pyramid cross-fusion of image and landmark streams with windowed attention. WHY: landmarks guide attention to expression-relevant regions (eyes, mouth) under pose/lighting change.

Mental Health Relevance: Candidate face encoder: strong RAF-DB/AffectNet performance with manageable size for real-time use.

[DAN] DAN: Distract Your Attention Network for Facial Expression Recognition (2023) — *WACV 2023*

Authors: Chuanhe Wen et al. | **Category:** Facial Expression Backbone
Agent Mentions (1/8): [Kimi]
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Occlusions and non-expressive regions distract FER attention maps.

Inputs / Outputs: In: face image. Out: expression class with multi-head attention.

Methodology (How & Why): HOW: multi-head cross-attention that is explicitly distracted away from irrelevant regions. WHY: robustness to partial occlusion (hands, glasses, masks).

Mental Health Relevance: Occlusion robustness matters for webcam use (hands near face, poor framing).

[ARM] Affective Representation Modeling (ARM) for Facial Expression Recognition (2022) — *CVPR 2022*

Authors: Various | **Category:** Facial Expression Backbone
Agent Mentions (0/8): *[Foundational Literature Baseline]*
Reading Priority: **[PRIORITY P3 - BACKGROUND]** | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: FER backbones need compact representations that transfer across in-the-wild conditions.

Inputs / Outputs: In: face images. Out: expression predictions.

Methodology (How & Why): HOW: representation-learning objectives tailored to affective features. WHY: compact affective embeddings fuse more cleanly with voice features downstream.

Mental Health Relevance: Background FER option for the face branch if POSTER++/DAN underperform on the custom set.

[OpenFace2] OpenFace 2.0: Facial Behavior Analysis Toolkit (2018) — *IEEE FG 2018*

Authors: Tadas Baltrusaitis et al. | **Category:** Vision Toolkit / AU Extractor
Agent Mentions (2/8): **[GLM]** **[Qwen]**
Reading Priority: **[PRIORITY P1 - IMPORTANT]** | **Links:** **[GitHub]** • [ArXiv Search] • [Scholar Search]

Problem Solved: Researchers need reproducible facial landmarks, action units, gaze and head pose without training their own detectors.

Inputs / Outputs: In: face video. Out: landmarks, AUs, gaze, pose.

Methodology (How & Why): HOW: engineered + learned pipeline (CE-CLM landmarks, AU classifiers). WHY: action units are psychologically grounded (FACS) features clinicians understand.

Mental Health Relevance: AU time-series are ideal inputs for explainable trend reports ('AU4 brow-lowering elevated all week').

[MediaPipe-Face] MediaPipe 468-point 3D Face Mesh and Blendshapes (2020) — *CVPR Workshops 2020*

Authors: Google Research | **Category:** Real-Time 3D Vision Toolkit
Agent Mentions (1/8): **[MiniMax]**
Reading Priority: **[PRIORITY P1 - IMPORTANT]** | **Links:** **[Docs]** • [ArXiv Search] • [Scholar Search]

Problem Solved: On-device real-time face tracking with minimal latency and no server round-trip.

Inputs / Outputs: In: camera frames. Out: 468 3D landmarks + blendshape coefficients.

Methodology (How & Why): HOW: lightweight cascaded networks optimized for mobile GPUs. WHY: sub-frame latency with on-device privacy (frames never leave the phone).

Mental Health Relevance: Default face front-end for the mobile app: private, real-time, and free.

[Wav2Vec2] wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations (2020) — *NeurIPS 2020*

Authors: Alexei Baevski et al. | **Category:** Acoustic Foundation Backbone
Agent Mentions (4/8): **[Claude]** **[DeepSeek]** **[Kimi]** **[MiniMax]**
Reading Priority: **[PRIORITY P0 - READ FIRST]** | **Links:** **[Paper]** • **[Toolkit]** • [ArXiv Search] • [Scholar Search]

Problem Solved: Labelled emotional speech is scarce; supervised SER overfits small corpora.

Inputs / Outputs: In: raw waveform. Out: contextualized speech representations.

Methodology (How & Why): HOW: masked contrastive pretraining on 50k+ hours of unlabeled speech, then light fine-tuning. WHY: self-supervision transfers across speakers and channels, cutting the speaker-independence gap.

Mental Health Relevance: Default voice encoder: fine-tune on RAVDESS/CREMA-D then the custom set; freeze-then-unfreeze schedule (LR ~2e-5) per DeepSeek findings.

[HuBERT] Hidden-Unit BERT for Speech (HuBERT) (2021) — *IEEE TASLP 2021*

Authors: Wei-Ning Hsu et al. | **Category:** Acoustic Foundation Backbone
Agent Mentions (6/8): [Claude] [DeepSeek] [Gemini] [GLM] [Kimi] [MiniMax]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [Paper] • [Toolkit] • [ArXiv Search] • [Scholar Search]

Problem Solved: Contrastive speech pretraining is sensitive to negative sampling and hyperparameters.

Inputs / Outputs: In: raw waveform. Out: discrete-unit BERT-style representations.

Methodology (How & Why): HOW: masked prediction of clustered acoustic units, iteratively refined. WHY: BERT-style objectives are stabler than contrastive ones and capture longer prosodic context.

Mental Health Relevance: Alternative voice encoder; compare against wav2vec 2.0 in ablations — prosody capture may favor HuBERT for distress cues.

[WavLM] WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing (2022) — *IEEE JSTSP 2022*

Authors: Sanyuan Chen et al. | **Category:** Acoustic Foundation Backbone
Agent Mentions (5/8): [Claude] [DeepSeek] [Gemini] [Kimi] [MiniMax]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [Paper] • [ArXiv Search] • [Scholar Search]

Problem Solved: One backbone should serve recognition, speaker and emotion tasks jointly (denoising + speaker awareness).

Inputs / Outputs: In: raw/noisy speech. Out: universal representations.

Methodology (How & Why): HOW: masked prediction with simulated noisy/overlapped speech during pretraining. WHY: built-in denoising matches the 'background noise' evaluation condition.

Mental Health Relevance: Best backbone when the deployment environment is noisy homes/streets rather than quiet rooms.

[Whisper] Robust Speech Recognition via Large-Scale Weak Supervision (Whisper) (2023) — *ICML 2023*

Authors: Alec Radford et al. | **Category:** ASR & Paralinguistic Backbone
Agent Mentions (5/8): [Claude] [Gemini] [Kimi] [MiniMax] [Qwen]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [Paper] • [Streaming] • [ArXiv Search] • [Scholar Search]

Problem Solved: The trimodal pipeline needs transcripts, but ASR errors propagate into fusion.

Inputs / Outputs: In: speech audio. Out: multilingual transcripts (+word timestamps via WhisperX).

Methodology (How & Why): HOW: encoder-decoder transformer trained on 680k hours of weakly supervised audio. WHY: robustness to accents, noise and disfluencies far beyond prior ASR.

Mental Health Relevance: Transcript stream for trimodal fusion; WhisperX word timestamps give utterance segmentation for the streaming pipeline.

[openSMILE-eGeMAPS] openSMILE and the Geneva Minimalistic Acoustic Parameter Set (eGeMAPS) (2016) — *IEEE TAFFC 2016*

Authors: Florian Eyben et al. | **Category:** Acoustic Handcrafted Features
Agent Mentions (3/8): [Claude] [GLM] [Kimi]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [Docs] • [ArXiv Search] • [Scholar Search]

Problem Solved: Deep features are opaque; clinicians want interpretable acoustic measures (pitch, jitter, shimmer).

Inputs / Outputs: In: speech waveform. Out: 88 eGeMAPS functionals (pitch, loudness, spectral, temporal).

Methodology (How & Why): HOW: curated minimal set of voice-science parameters with open-source extraction. WHY: interpretability + tiny compute + a like-for-like baseline against SSL embeddings.

Mental Health Relevance: Use eGeMAPS as the interpretable voice baseline (thesis requirement: MFCC/pitch/spectral) and surface pitch/energy trends in the dashboard.

[DAIC-WOZ-Paper] The Distress Analysis Interview Corpus (DAIC): Architecture and Pre-Validation (2014) — *LREC 2014*

Authors: Jonathan Gratch et al. | **Category:** Clinical Mental Health Corpus
Agent Mentions (3/8): [Claude] [Gemini] [MiniMax]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [Dataset] • [ArXiv Search] • [Scholar Search]

Problem Solved: No synchronized audiovisual corpus linked depression/distress labels to observable behavior.

Inputs / Outputs: In: clinical interviews (audio, video, transcripts). Out: PHQ-8 depression scores.

Methodology (How & Why): HOW: Wizard-of-Oz virtual-interviewer protocol with standardized depression screening. WHY: controlled elicitation plus validated clinical scales bridges basic emotions to mental-health states.

Mental Health Relevance: The bridge dataset: pretrain/fine-tune here to transfer basic-emotion features toward depression-relevant representations.

[DepAudioNet] DepAudioNet: An Efficient Deep Model for Audio-based Depression Classification (2016) — *AVEC 2016*

Authors: Ma et al. | **Category:** Acoustic Depression Detection
Agent Mentions (0/8): *[Foundational Literature Baseline]*
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Depression must be detectable from voice alone, efficiently, on AVEC data.

Inputs / Outputs: In: interview audio (mel-spectrograms). Out: depressed / non-depressed.

Methodology (How & Why): HOW: compact CNN+LSTM over mel features. WHY: proves a small model can capture psychomotor-slowing cues (flat prosody, long pauses).

Mental Health Relevance: Reference voice-only depression baseline; pause-duration and speech-rate features transfer to the monitoring app.

[Deep-Multimodal-Depression] Deep Multimodal Analysis for Depression Prediction and Severity Estimation (2018) — *CVPR 2018*

Authors: Yang et al. | **Category:** Multimodal Depression Detection
Agent Mentions (0/8): *[Foundational Literature Baseline]*
Reading Priority: [PRIORITY P2 - SPECIALIZED] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Depression severity estimation needs joint audio+visual modelling, not just classification.

Inputs / Outputs: In: interview audio+video. Out: depression presence + severity score.

Methodology (How & Why): HOW: deep multimodal regression over audiovisual behavior descriptors. WHY: severity scores (not binary labels) match clinical use.

Mental Health Relevance: Precedent for outputting a continuous wellness score alongside categorical emotion — exactly the dashboard's trend line.

[AVEC-Challenges] AVEC 2019 Workshop and Challenge: State-of-Mind, Detecting Depression with AI, and Cross-Cultural Affect (2019) — *ACM MM 2019*

Authors: Fabien Ringeval et al. | **Category:** Affective Challenge & Benchmark
Agent Mentions (3/8): [Claude] [Gemini] [MiniMax]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Field needs shared depression/affect tasks with fixed protocols to compare methods.

Inputs / Outputs: In: challenge audiovisual data. Out: ranked depression/affect predictions.

Methodology (How & Why): HOW: standardized sub-challenges (depression severity, affect dimensions) with hidden test labels. WHY: prevents split-hacking and overfitting to friendly partitions.

Mental Health Relevance: Adopt AVEC-style hidden-split discipline for the custom dataset evaluation to keep claims credible.

[Mental-Health-Distress-Review] A Systematic Review on Automated Multimodal Mental Health and Distress Detection (2026) — *Springer Nature 2026*

Authors: Various | **Category:** Clinical Application Survey
Agent Mentions (6/8): [ChatGPT] [Claude] [DeepSeek] [Gemini] [Kimi] [MiniMax]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [Paper] • [ArXiv Search] • [Scholar Search]

Problem Solved: No consolidated view of how MER techniques transfer to clinical distress detection with privacy constraints.

Inputs / Outputs: In: 100+ clinical MER studies. Out: taxonomy + privacy/personalization recommendations.

Methodology (How & Why): HOW: systematic review across sensors, models and deployment studies. WHY: identifies federated learning, differential privacy and on-device inference as the viable privacy stack.

Mental Health Relevance: Primary citation for the privacy chapter and the 'no complete privacy-preserving MER pipeline exists' gap claim.

[Rouast-Survey] Deep Learning for Multimodal Emotion Recognition: A Review (2021) — *IEEE TAFFC 2021*

Authors: Rouast et al. | **Category:** Landscape Survey
Agent Mentions (0/8): *[Foundational Literature Baseline]*
Reading Priority: [PRIORITY P3 - BACKGROUND] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Early/late/hybrid fusion options needed a unified comparison framework.

Inputs / Outputs: In: pre-2021 MER literature. Out: fusion taxonomy + best-practice guidance.

Methodology (How & Why): HOW: structured comparison of fusion strategies and modality contributions. WHY: established the early/late/hybrid vocabulary still used in theses.

Mental Health Relevance: Background citation for the thesis Chapter 2 fusion taxonomy.

[Gandhi-Survey] Multimodal Sentiment Analysis: A Systematic Review of History, Datasets and Models (2023) — *Information Fusion 2023*

Authors: Gandhi et al. | **Category:** Landscape Survey
Agent Mentions (0/8): *[Foundational Literature Baseline]*
Reading Priority: [PRIORITY P3 - BACKGROUND] | **Links:** [Paper] • [ArXiv Search] • [Scholar Search]

Problem Solved: MSA history, datasets and models had never been systematically unified.

Inputs / Outputs: In: MSA literature. Out: historical taxonomy + dataset catalogue.

Methodology (How & Why): HOW: systematic review with dataset/model matrices. WHY: dataset catalogue directly feeds corpus selection.

Mental Health Relevance: Dataset-selection reference for the thesis; cross-check corpus metadata against it.

[Baltrusaitis-Survey] Multimodal Machine Learning: A Survey and Taxonomy (2018) — *IEEE TPAMI 2018*

Authors: Baltrusaitis et al. | **Category:** Foundational Taxonomy Survey
Agent Mentions (0/8): *[Foundational Literature Baseline]*
Reading Priority: [PRIORITY P3 - BACKGROUND] | **Links:** [Paper] • [ArXiv Search] • [Scholar Search]

Problem Solved: Multimodal ML lacked shared vocabulary (representation, translation, alignment, fusion, co-learning).

Inputs / Outputs: In: multimodal ML literature. Out: five-challenge taxonomy.

Methodology (How & Why): HOW: defines representation/translation/alignment/fusion/co-learning challenges. WHY: every later MER survey builds on this framing.

Mental Health Relevance: Foundational framing citation for the thesis introduction.

[Schuller-Affect-Survey] A Review on Recent Affect Recognition Challenges: Methods and Directions (2021) — *Proceedings of the IEEE 2021*

Authors: Bjorn Schuller et al. | **Category:** Benchmark Survey
Agent Mentions (0/8): *[Foundational Literature Baseline]*
Reading Priority: [PRIORITY P3 - BACKGROUND] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Challenge-driven progress (AVEC, ABAW, ComParE) needed synthesis into durable lessons.

Inputs / Outputs: In: recent affect challenges. Out: methods/directions synthesis.

Methodology (How & Why): HOW: meta-analysis of winning challenge systems. WHY: winning recipes (ensembles, SSL backbones) generalize.

Mental Health Relevance: Justifies the evaluation protocol: challenges show speaker-independent hidden splits are the honest metric.

[MER-Mega-Survey-2026] A Survey on Multimodal Emotion Recognition: Methods, Datasets, and Future Directions (2026) — *Information Fusion / ScienceDirect 2026*

Authors: Various | **Category:** Comprehensive Modern Survey
Agent Mentions (2/8): [Claude] [DeepSeek]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [Paper] • [ArXiv Search] • [Scholar Search]

Problem Solved: No current survey covers the transformer/MLLM-era fusion landscape end to end.

Inputs / Outputs: In: modern MER literature. Out: early/late/hybrid + attention/graph fusion taxonomy.

Methodology (How & Why): HOW: categorizes fusion into early, late and hybrid with dynamic-weighting and cross-modal-attention variants. WHY: declares 'effective fusion design remains open' — simple concatenation/voting under-exploits complementarity.

Mental Health Relevance: Backbone citation for choosing attention-based selective fusion over concatenation in the blueprint.

[SER-Mega-Review-2026] Speech Emotion Recognition: A Comprehensive Survey of Modern Architectures (2026) — *Information Fusion 2026*

Authors: Various | **Category:** Acoustic Survey
Agent Mentions (1/8): [DeepSeek]
Reading Priority: [PRIORITY P0 - READ FIRST] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: Eleven open SER questions (Representations? generalization? robustness?) needed joint answers.

Inputs / Outputs: In: modern SER literature. Out: 11-question synthesis; SSL (wav2vec2/HuBERT) declared SOTA.

Methodology (How & Why): HOW: systematic mega-review of techniques and pipelines. WHY: establishes SSL representations + speaker-independence as the deployment baseline.

Mental Health Relevance: Justifies replacing the MFCC+LSTM voice plan with SSL encoders (keeping MFCC only as interpretable baseline).

[FER-Survey-2026] Facial Expression Recognition in the Wild: A Contemporary Survey (2025/2026) — *IEEE TPAMI / Neurocomputing 2025*

Authors: Various | **Category:** Facial Expression Survey
Agent Mentions (3/8): [ChatGPT] [Claude] [DeepSeek]
Reading Priority: [PRIORITY P1 - IMPORTANT] | **Links:** [ArXiv Search] • [Scholar Search]

Problem Solved: FER moved from handcrafted features to ViTs and micro-expression modelling; the story needed retelling.

Inputs / Outputs: In: FER literature (CNN to ViT era). Out: architecture evolution + temporal/micro-expression trends.

Methodology (How & Why): HOW: traces handcrafted-to-end-to-end evolution with dataset/robustness analysis. WHY: documents why ViT + temporal modelling now beats frame CNNs.

Mental Health Relevance: Supports the ViT face encoder choice and the micro-expression future-work item.

Comprehensive Multimodal Datasets Directory (15 Corpora)

Complete directory of multimodal emotional and clinical mental health datasets referenced across all reviews:

IEMOCAP (2008) — *Text, Video, Audio*

Size / Scale: 12h, 10 actors, ~10k utterances | **Annotations:** 9 emotions + valence/arousal/dominance
License: Research license on request | **Access URL:** Request via https://sail.usc.edu/iemocap/
Agent Mentions (8/8): [ChatGPT] [Claude] [DeepSeek] [Gemini] [GLM] [Kimi] [MiniMax] [Qwen]

Mental Health Usage & Research Discipline: Primary dyadic-conversation benchmark. Pretrain fusion here, but NEVER report it as mental-health evidence (acted English).

MELD (2019) — *Text, Video, Audio*

Size / Scale: ~13k utterances (Friends TV) | **Annotations:** 7 emotions + sentiment
License: Research use | **Access URL:** https://affective-meld.github.io/
Agent Mentions (8/8): [ChatGPT] [Claude] [DeepSeek] [Gemini] [GLM] [Kimi] [MiniMax] [Qwen]

Mental Health Usage & Research Discipline: Multi-party conversation benchmark for DialogueRNN/GCN/COSMIC-style context models. Scripted-TV bias must be disclosed.

CMU-MOSI / CMU-MOSEI (2017/2018) — *Text, Video, Audio*

Size / Scale: MOSI 2.2k clips; MOSEI 23.5k clips | **Annotations:** Sentiment intensity [-3,3] + emotions (MOSEI)
License: Research use | **Access URL:** https://github.com/CMU-MultiComp-Lab/CMU-MultimodalSDK
Agent Mentions (7/8): [ChatGPT] [Claude] [DeepSeek] [Gemini] [GLM] [MiniMax] [Qwen]

Mental Health Usage & Research Discipline: Standard fusion-benchmark pair (MulT, MISA, Self-MM numbers). Opinion-video domain; report MAE/Acc-7/Acc-2/F1 exactly as MERBench prescribes.

RAVDESS (2018) — *Audio, Video (speech+song)*

Size / Scale: 24 actors x 60 clips | **Annotations:** 8 emotions, 2 intensities
License: Research use | **Access URL:** https://www.innovatiana.com/en/datasets/ravdess
Agent Mentions (7/8): [ChatGPT] [Claude] [DeepSeek] [Gemini] [Kimi] [MiniMax] [Qwen]

Mental Health Usage & Research Discipline: Project-named corpus: fine-tune voice encoder here. Acted and clean — always pair with speaker-independent splits (expect ~13% drop vs random splits).

CREMA-D (2014) — *Audio, Video*

Size / Scale: 91 actors, 7,442 clips | **Annotations:** 6 emotions, 4 intensities
License: Research use | **Access URL:** https://github.com/CheyneyComputerScience/CREMA-D
Agent Mentions (5/8): [ChatGPT] [Claude] [DeepSeek] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Project-named corpus: diversity (age/ethnicity) makes it the fairer voice-evaluation set. Combine with RAVDESS for SER training.

FER2013 (2013) — *Static face images (48x48 gray)*

Size / Scale: 35,887 images | **Annotations:** 7 basic emotions
License: Public (Kaggle) | **Access URL:** https://www.kaggle.com/datasets/msambare/fer2013
Agent Mentions (5/8): [ChatGPT] [Claude] [DeepSeek] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Project-named corpus: fine-tune ViT face encoder (expect ~70% acc). Heavy label noise + imbalance — use class-weighted loss; report per-class F1.

AffectNet (2017) — *Static face images (in-the-wild)*

Size / Scale: ~1M images (450k annotated) | **Annotations:** 8 emotions + valence/arousal
License: Research request | **Access URL:** http://mohammadmahoor.com/affectnet/
Agent Mentions (5/8): [ChatGPT] [Claude] [DeepSeek] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Project-named corpus: in-the-wild pretraining for the face encoder; valence/arousal heads seed the dimensional trend output.

DAIC-WOZ / E-DAIC (2014/2019) — *Audio, Video, Text + PHQ-8*

Size / Scale: ~190 interviews | **Annotations:** PHQ-8 depression scores
License: Restricted research agreement | **Access URL:** https://dcapswoz.ict.usc.edu/
Agent Mentions (3/8): [Claude] [Gemini] [MiniMax]

Mental Health Usage & Research Discipline: Clinical bridge: the ONLY listed corpus with validated depression labels. Fine-tune here to transfer basic-emotion features toward distress detection.

ABAW / Aff-Wild2 (2019-2024) — *Face video in-the-wild*

Size / Scale: ~600 videos, ~3M frames | **Annotations:** VA + AUs + 8 expressions
License: Challenge agreement | **Access URL:** https://affective-behavior-analysis-in-the-wild.github.io/8th/
Agent Mentions (2/8): [Claude] [MiniMax]

Mental Health Usage & Research Discipline: Dimensional (valence-arousal) + AU supervision for the trend head; competition protocol is the robustness gold standard.

DFEW (2021) — *Video clips in-the-wild*

Size / Scale: ~16k clips | **Annotations:** 7 emotions
License: Research use | **Access URL:** https://dfew-dataset.github.io/
Agent Mentions (5/8): [ChatGPT] [Claude] [Gemini] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Dynamic FER benchmark for the video face encoder (temporal modelling beyond static FER2013).

MER2023 / MER2024 Challenges (2023/2024) — *Audio, Video, Text*

Size / Scale: Challenge-scale unlabeled + labeled | **Annotations:** Discrete + dimensional + open-vocabulary
License: Challenge terms | **Access URL:** https://github.com/zeroQiaoba/MERTools/tree/master/MER2024
Agent Mentions (6/8): [ChatGPT] [Claude] [Gemini] [GLM] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Adopt MER2024 tracks (incl. open-vocabulary reasoning) for the explainable-head evaluation.

GoEmotions (2020) — *Text (Reddit comments)*

Size / Scale: 58k comments | **Annotations:** 27 emotions + neutral
License: Apache-2.0 | **Access URL:** https://github.com/google-research/google-research/tree/master/goemotions
Agent Mentions (2/8): [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Fine-grained text-emotion labels for the transcript stream; maps basic emotions toward nuanced states.

RAF-DB (2017) — *Static face images*

Size / Scale: ~30k images | **Annotations:** 7 basic + 12 compound emotions
License: Research use | **Access URL:** Research request (W. Deng lab)
Agent Mentions (3/8): [Claude] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Cleaner FER2013 alternative for face-encoder benchmarking (POSTER++/DAN numbers reported here).

MSP-Podcast / MSP-IMPROV (2017/2025) — *Audio (+video for IMPROV)*

Size / Scale: 100k+ turns (Podcast) | **Annotations:** Categorical + dimensional
License: Research request | **Access URL:** https://lab-msp.com/MSP/
Agent Mentions (1/8): [Qwen]

Mental Health Usage & Research Discipline: Naturalistic (non-acted) speech-emotion data — closer to monitoring reality than RAVDESS; prioritise for voice generalization tests.

CH-SIMS / SIMS (2020) — *Text, Audio, Video (Chinese)*

Size / Scale: ~2.3k clips | **Annotations:** Multimodal + unimodal sentiment
License: Research use | **Access URL:** Academic request
Agent Mentions (2/8): [Claude] [MiniMax]

Mental Health Usage & Research Discipline: Methodology reference only (unimodal annotation scheme worth copying for the custom dataset); excluded from English benchmarks by the MiniMax filter.

Open-Source Toolkits, Frameworks & Implementation Codebases (13 Tools)

Production-grade toolkits and repositories recommended for building the mental health monitoring pipeline:

M-SENA platform

Functionality: Integrated MSA/MER platform: data, features, models, evaluation.
Repository / Docs: ACL 2023 system demo (search: M-SENA multimodal sentiment platform)
Agent Mentions (1/8): [GLM]

Role in Proposed System: Experiment harness for face-only / voice-only / fused ablations.

MERTools / MERBench

Functionality: Unified MER data pipeline, fixed splits, standard metrics + challenge toolkit.
Repository / Docs: https://github.com/zeroQiaoba/MERTools
Agent Mentions (2/8): [ChatGPT] [MiniMax]

Role in Proposed System: Adopt its protocol so thesis numbers are comparable across corpora.

CMU-MultimodalSDK

Functionality: Reference SDK for MOSI/MOSEI loading and classic baselines (TFN, MFN, MulT).
Repository / Docs: https://github.com/CMU-MultiComp-Lab/CMU-MultimodalSDK
Agent Mentions (1/8): [MiniMax]

Role in Proposed System: Reproduce baseline fusion numbers before claiming improvements.

S3PRL / SUPERB

Functionality: Self-supervised speech toolkit (wav2vec2, HuBERT, WavLM recipes).
Repository / Docs: https://github.com/s3prl/s3prl
Agent Mentions (1/8): [Claude]

Role in Proposed System: Voice-encoder fine-tuning recipes with frozen-layer schedules.

SpeechBrain

Functionality: Speech toolkit incl. emotion-recognition recipes.
Repository / Docs: https://github.com/speechbrain/speechbrain
Agent Mentions (1/8): [Qwen]

Role in Proposed System: Alternative SER training harness; strong recipe collection.

WhisperX

Functionality: Whisper ASR + word-level timestamps + diarization.
Repository / Docs: https://github.com/m-bain/whisperx
Agent Mentions (2/8): [Gemini] [Qwen]

Role in Proposed System: Transcript stream + utterance segmentation for the streaming pipeline.

FunASR

Functionality: Industrial ASR toolkit (paraformer) with punctuation/timestamps.
Repository / Docs: https://github.com/modelscope/FunASR
Agent Mentions (1/8): [Qwen]

Role in Proposed System: Low-latency ASR alternative for the real-time head.

openSMILE / eGeMAPS

Functionality: Interpretable acoustic feature extractor (88-function eGeMAPS set).
Repository / Docs: https://audeering.github.io/opensmile/
Agent Mentions (3/8): [Claude] [GLM] [Kimi]

Role in Proposed System: MFCC/pitch/spectral baseline features (thesis requirement) + dashboard-grade interpretable trends.

OpenFace 2.0

Functionality: Landmarks, action units, gaze, head pose.
Repository / Docs: https://github.com/TadasBaltrusaitis/OpenFace
Agent Mentions (2/8): [GLM] [Qwen]

Role in Proposed System: AU time-series for explainable reports.

MediaPipe Face

Functionality: 468-point 3D mesh + blendshapes, on-device.
Repository / Docs: https://ai.google.dev/edge/mediapipe
Agent Mentions (1/8): [MiniMax]

Role in Proposed System: Default mobile face front-end (private, real-time).

FER backbones (POSTER / EfficientFace / MAE-DFER / DDAMFN / FaceChannel)

Functionality: Open FER implementations spanning CNN, transformer and masked-autoencoder designs.
Repository / Docs: https://github.com/zczcwh/POSTER ; https://github.com/zengqunzhao/EfficientFace ; https://github.com/sunlicai/MAE-DFER ; https://github.com/SainingZhang/DDAMFN ; https://github.com/pablovin/FaceChannel
Agent Mentions (2/8): [Claude] [MiniMax]

Role in Proposed System: Face-encoder shortlist for ablations.

Conversation-ERC codebases (conv-emotion / COGMEN / MMGCN)

Functionality: Reference DialogueRNN/COSMIC, COGMEN and MMGCN implementations.
Repository / Docs: https://github.com/declare-lab/conv-emotion ; https://github.com/exploration-lab/COGMEN ; https://github.com/hujingwen6666/MMGCN
Agent Mentions (2/8): [Claude] [MiniMax]

Role in Proposed System: Context-modelling references for the session-level mood GRU.

Real-time demo repos

Functionality: mrw-soumik real-time face+voice pipeline; ShahnawazKakarh SSL transfer-learning SER (RAVDESS WF1 0.858 random / 0.728 speaker-independent).
Repository / Docs: GitHub search: mrw-soumik Multimodal-Emotion-Recognition-System ; ShahnawazKakarh speech-emotion-recognition-transfer-learning
Agent Mentions (1/8): **[DeepSeek]**

Role in Proposed System: Starting skeleton for the streaming app + the speaker-independence warning (13% drop) that shapes evaluation.

Key Research Challenges & Critical Gaps (C1 — C8)

Eight unresolved scientific challenges identified across the multi-agent synthesis:

C1. Single-user continuous dynamics (speaking/silent transitions)

Raised by Agents (3/8): [DeepSeek] [ChatGPT] [MiniMax]

Analysis & Mitigation: MCER literature models multi-speaker influence; the project has ONE user alternating between speaking (trimodal) and silent (face-only) states. No benchmark covers this transition modelling.

C2. Missing-modality robustness under streaming

Raised by Agents (5/8): [ChatGPT] [DeepSeek] [Kimi] [MiniMax] [Qwen]

Analysis & Mitigation: Camera-off, mic-muted, or dropped streams must not crash inference. Diffusion/latent reconstruction (HVDER ~75% at 70% missing) is unvalidated in real-time; MissBench shows shared-rate robustness hides inequity under imbalanced missingness.

C3. Speaker independence & environment generalization

Raised by Agents (3/8): [DeepSeek] [Claude] [Gemini]

Analysis & Mitigation: ~13% WF1 drop from random to speaker-independent splits; lighting, noise, accents and speaking styles compound it. Uncontrolled-condition evaluation must be cross-speaker by construction.

C4. Privacy-preserving MER pipeline

Raised by Agents (4/8): [DeepSeek] [Claude] [Gemini] [ChatGPT]

Analysis & Mitigation: Faces + voices are biometric. Federated learning, differential privacy and on-device inference are recommended everywhere but no complete privacy-preserving MER pipeline exists.

C5. Basic emotions to mental-health states

Raised by Agents (3/8): [DeepSeek] [Claude] [Qwen]

Analysis & Mitigation: Benchmarks output happy/sad/angry; monitoring needs valence-arousal trends and clinical-scale mapping (PHQ-8 via DAIC-WOZ). The categorical-to-dimensional translation layer is understudied.

C6. Calibration & trustworthy confidence

Raised by Agents (4/8): [ChatGPT] [Claude] [DeepSeek] [Gemini]

Analysis & Mitigation: Overconfident models mislead users about their own state. Calibrated confidence (MER-Calibration) is a safety requirement for the dashboard scores.

C7. Asynchronous alignment & the speech-articulation paradox

Raised by Agents (2/8): [Gemini] [Qwen]

Analysis & Mitigation: Modalities are asynchronous; worse, speaking itself deforms the lower face, confounding FER. Alignment modules (TITAN-style gating) and articulation-aware face modelling are required.

C8. No real-mental-health conversational dataset

Raised by Agents (3/8): [DeepSeek] [MiniMax] [Claude]

Analysis & Mitigation: IEMOCAP/MOSEI/MELD are English acted/media data. No corpus captures real mental-health conversations with synced face+voice+language — the custom dataset is the most publishable contribution.

End-to-End System Architecture Blueprint (Mapped to idea.docx)

The blueprint realises the idea.docx workflow (face analysis + voice analysis + multimodal fusion + visualization) as a dual-head system: HEAD-A (discriminative) outputs calibrated emotion + dimensional scores under a streaming latency budget; HEAD-B (explainable MLLM, Emotion-LLaMA/AffectGPT pattern) grounds short rationales in HEAD-A evidence. Edge-first privacy keeps raw biometrics on device; only embeddings or aggregates ever leave it.

Subsystem Component	Architectural Implementation & idea.docx Mapping
Ingestion & segmentation	Camera frames + mic audio -> WhisperX utterance segmentation with 3-s rolling audio windows; silent gaps routed to face-only path (C1). Maps to idea.docx Input + Processing steps 1-2.
Face branch	MediaPipe mesh (mobile) or POSTER++ encoder; OpenFace AUs for explainability; AffectNet pretrain -> FER2013 tune -> custom-set tune. Covers idea.docx facial analysis + robustness under lighting/variation.
Voice branch	Wav2Vec2/HuBERT encoder (primary) + eGeMAPS/MFCC interpretable baseline (thesis requirement); RAVDESS+CREMA-D fine-tuning with speaker-independent splits. Covers idea.docx voice analysis.
Fusion core (HEAD-A)	MulT-style cross-modal attention with MISA disentanglement + M3ER reliability gating + MMIN/M3S missing-modality training; calibration layer (MER-Calibration) before any score is shown. Covers idea.docx multimodal fusion.
Explainable head (HEAD-B)	Instruction-tuned MLLM (Emotion-LLaMA pattern) generating 1-2 sentence rationales citing HEAD-A evidence (AU4, pitch drop, hotspot timestamp); evaluated on AffectGPT-style reasoning checks.
Trend & visualization layer	Valence-arousal + categorical history, calibrated confidence bars, hotspot markers (Centering-Hotspots), PHQ-8-adjacent wellness scale via DAIC-WOZ transfer. Covers idea.docx visualization + dashboard.
Privacy & edge envelope	On-device inference (MediaPipe/CubeMLP-class fusion), federated updates, differential-privacy aggregates; raw video/voice never uploaded. Covers idea.docx privacy-aware prototype (C4).

Streaming Latency Budget & Real-Time Constraints

Pipeline Stage	Target Budget	Implementation Engine & Constraints
Face detect+encode (per frame)	<= 40 ms	MediaPipe / POSTER++-small
Voice window classify (3-s roll)	<= 300 ms	Wav2Vec2-base distilled / CubeMLP
Cross-modal fusion + calibration	<= 120 ms	Distilled MulT / LMF fallback
ASR partial (streaming)	<= 500 ms	Whisper-streaming / FunASR
Explanation (async, non-blocking)	<= 2 s	MLLM head off critical path
End-to-end utterance verdict	<= 1 s	Heads A+B decoupled; B async

Master Experimental Evaluation Matrix

Evaluation Objective	Benchmark Datasets	Target Metrics & Verification Discipline
Face-only / voice-only / fused comparison	FER2013, AffectNet, RAVDESS, CREMA-D	Acc, macro-F1, per-class F1, confusion matrix
Speaker independence	Leave-speakers-out splits	WF1 gap vs random split must be reported (DeepSeek 13% warning)
Missing-modality matrix	Camera-off / mic-off / intermittent (MissBench MEI/MLI)	Acc@missing-rate curve; honest 'missing' flags
Real-world robustness	Custom set: lighting x noise x speaking style	Stratified F1 + latency percentiles
Calibration	Reliability diagrams, ECE	ECE < 0.05 before dashboard release
Reasoning quality	AffectGPT-style grounding checks	Faithfulness / evidence-citation rate